

Victoria Junior College

JC2 Preliminary Examination 2009

9747 H2 Biology Paper 2

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Wednesday

1400 – 1600 h

16 September 2009

2 Hours

Name: _____

CT: _____

Instructions

1. This paper is divided into 2 sections.

Section A: 6 Structured questions

Section B: 2 Essay questions

2. **Answer ALL questions in Section A and ONE question in Section B.**

3. For Section A, answers must be written on the spaces provided and nowhere else.

For Section B, answers must be written on the writing paper.

Question No.	Marks
1	
2	
3	
4	
5	
6	
7 OR 8	Essay
Total	

Section A: Answer all questions.

Write your answers in the space provided and nowhere else.

Question 1 (12 marks)

Collagen is the most abundant protein in animals. The basic subunit of collagen is the α -chains which consist predominantly of repeating Gly-X-Y tripeptide motifs. (X and Y represent two different amino acids.) The presence of glycine at every third amino acid is essential to allow the α -chains to adopt the characteristic collagen triple helix. The triple helices can further associate to form fibers with high tensile strength. Collagen molecule is the basic building block of bone and tendon.

(a) Explain the importance of glycine in the formation of collagen triplex helix. [2]

(b) Describe how association between triple helices allows the formation of fibers with high tensile strength. [3]

Genetic mutations in the gene that codes for α -chain of collagen results in the formation of weak tendons and fragile bones characteristic of osteogenesis imperfecta or brittle bone disease. These mutations include gene mutations and RNA splicing mutations. A single base substitution can convert a codon for glycine to a codon for an amino acid with a bulkier side chain. But gene mutations do not always produce mutated collagen.

(c) Distinguish between gene mutations and RNA splicing mutations. [2]

(d) Explain why gene mutations do not always produce mutated collagen whereas mutations involving RNA splicing will always produce mutated collagen. [3]

In the biosynthesis of collagen, any gene mutation can have a direct effect on its structure and hence its function. Like collagen, cellulose serves a structural function. But unlike collagen, any gene mutation does not have a **direct** effect on the cellulose structure.

(e) Explain why. [2]

Question 2 (13 marks)

Human DNA has been analyzed and details of certain genes are shown in the table below.

Gene	Gene size / kb*	mRNA size / kb	Number of introns
Insulin	1.7	0.4	2
Collagen	38.0	5.0	50
Albumin	25.0	2.1	14
Phenylalanine hydroxylase	90.0	2.4	12
Dystrophin	2000.0	17.0	50

*Kilobase pairs

(Source: William S Klug and Michael R Cummings, (2002), Concepts of Genetics, 7th Edition, Prentice Hall, page 314)

(a) (i) Calculate the average size of the introns for the albumin gene. (Show your workings.) [1]

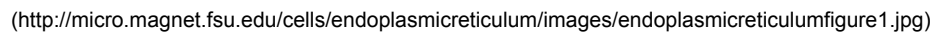
(ii) Analyze the relationship(s) between gene size and the number of introns. [2]

- (iii) Determine the maximum number of amino acids that could be produced by translating the phenylalanine hydroxylase mRNA. (Show your workings.) [1]

The DNA in each chromosome constitutes many genes as well as vast stretches of non-coding DNA, the function of which is unknown. A gene is any given segment along the DNA that encodes instructions that allow a cell to produce a specific product (e.g. enzyme).

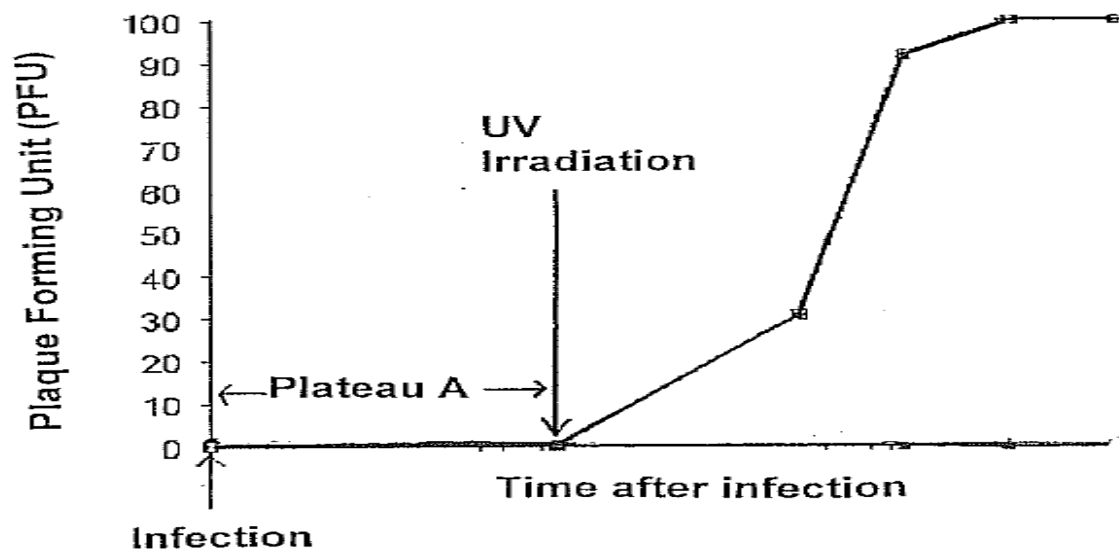
- (b) Describe how a eukaryotic cell turns on its genes in its chromatin fibres. [2]

- (c) State **three** differences between transcription and replication. [3]



Question 3 (14 Marks)

Lambda (λ) phage infections result in the lysis of the bacterial host cells which shows up as a plaque or a zone of clearing on an agar plate containing a dense growth of bacteria. This observation can be used to quantify the number of infectious bacteriophage particles through a plaque assay. 'Plaque Forming Unit (PFU)' gives the measurement of the number of infectious viral particles present during the experiment. The graph below shows the changes in PFU caused by lambda (λ) phages infecting *E.coli* cells upon irradiation by UV.



(a) Briefly describe the structure of a lambda (λ) phage. [3]

(b) Suggest an explanation for the flat part of the graph, labelled A, as shown above. [2]

(c) Account for the rise in PFU shortly after UV irradiation. [3]

(d) Describe the differences in the life cycle of T4 phage and HIV virus. [4]

(e) Suggest how a viral infection may cause the immune system to turn against its own body cells. [2]

Question 4 (14 marks)

The inheritance of flower length in tobacco plants is an example of polygenic inheritance. Plants from a group homozygous for all the short-flower alleles were crossed with plants from a group homozygous for all the long-flower alleles. The offspring from this cross (the F₁) were then self-fertilized to produce the next generation (F₂). Figure 4.1 shows the flower lengths in each of these groups of plants.

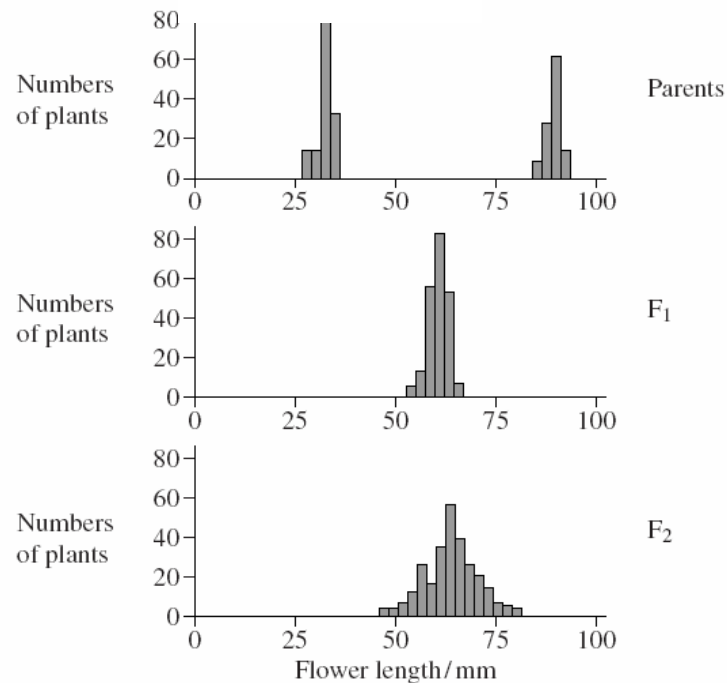


Figure 4.1. Flower Lengths in Each Group of Plants

(a) Describe **two** differences between inheritance caused by polygenes and multiple alleles. [2]

(b) Explain how the data for the parents and F1 support the fact that the inheritance of flower length is polygenic. [2]

(c) Account clearly for the difference in variation shown by F1 and F2. [4]

(d) Assuming that the plant is **diploid**, how would the results for F2 differ if flower length is caused by one gene with two alleles showing codominance relationship? [3]

(e) When comparing variation in polygenic characters between two groups of organisms, it is often more useful to compare standard deviations rather than ranges. Explain why. [2]

Question 5 (13 marks)

The cardiac action potential is a specialized action potential in the heart, with unique properties necessary for the conduction of electrical conduction system of the heart which involves muscle cells of the heart. The heart muscle cells share many things in common with nerve cells. Figure 5.1 below shows a schematic diagram of an action potential in the heart which can be divided into 5 phases: phase 0, phase 1, phase 2, phase 3 and phase 4. Each of which is characterized with their respective changes in membrane potential.

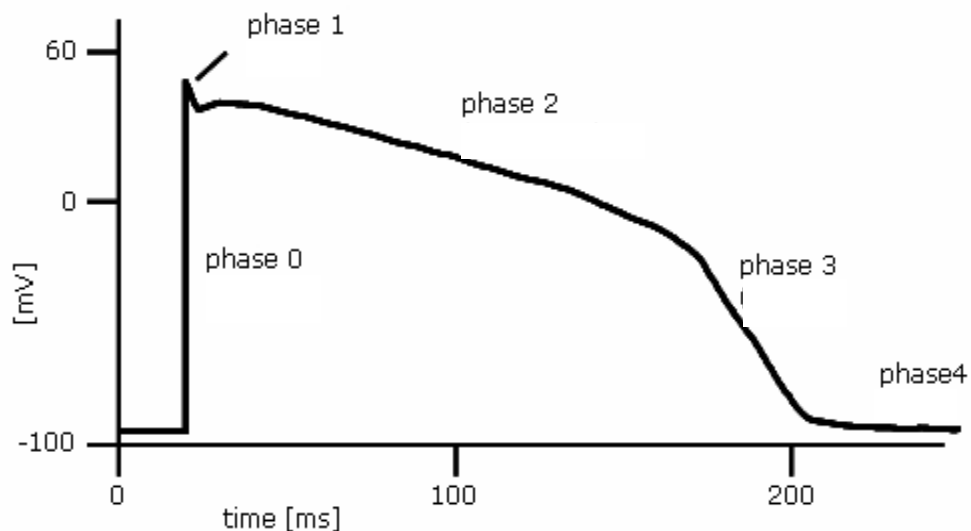


Figure 5.1. Various Phases of a Typical Cardiac Action Potential.

Table 5.2 below shows the relative concentration of ions inside and outside of the cardiac muscle cells.

Intra- and extracellular ion concentrations (mmol/L)				
Element	Ion	Extracellular	Intracellular	Ratio
Sodium	Na ⁺	135 – 145	10	14:1
Potassium	K ⁺	3.5 – 5.0	155	1:30
Chloride	Cl ⁻	95 – 110	20 – 30	4: 1
Calcium	Ca ²⁺	2	10 ⁻⁴	2 X 10 ⁴ : 1

Table 5.2. Relative Concentration of Ions Inside and Outside of a Heart Cell

(a) (i) State the resting potential for the cardiac action potential shown. [1]

(ii) With reference to the information shown in Table 5.2, suggest how the resting potential can be maintained in the heart muscles using sodium and potassium ions. [3]

(b) Using the information given in Figure 5.1, briefly explain the changes in Phase 0. [3]

(c) (i) Describe a difference at Phase 2 between a nerve cell and a cardiac heart cell. [1]

(ii) Phase 2 in the cardiac potential is known as the 'plateau' phase and it is caused by the inward flow of Ca^{2+} . Suggest how this may explain for the difference in (i). [2]

(iii) Cardiac arrhythmia is term used for any abnormal electrical activity in the heart. As a result the heart may beat too fast. Calcium channel blockers such as Verapamil are often used to treat this condition. Suggest the action of Verapamil in controlling this symptom of arrhythmia. [2]

(d) Draw into Figure 5.1 the duration of the 'refractory period'. [1]

Question 6 (14 marks)

Figure 6.1 below shows the classification of some primates. This classification is hierarchical and phylogenetic.

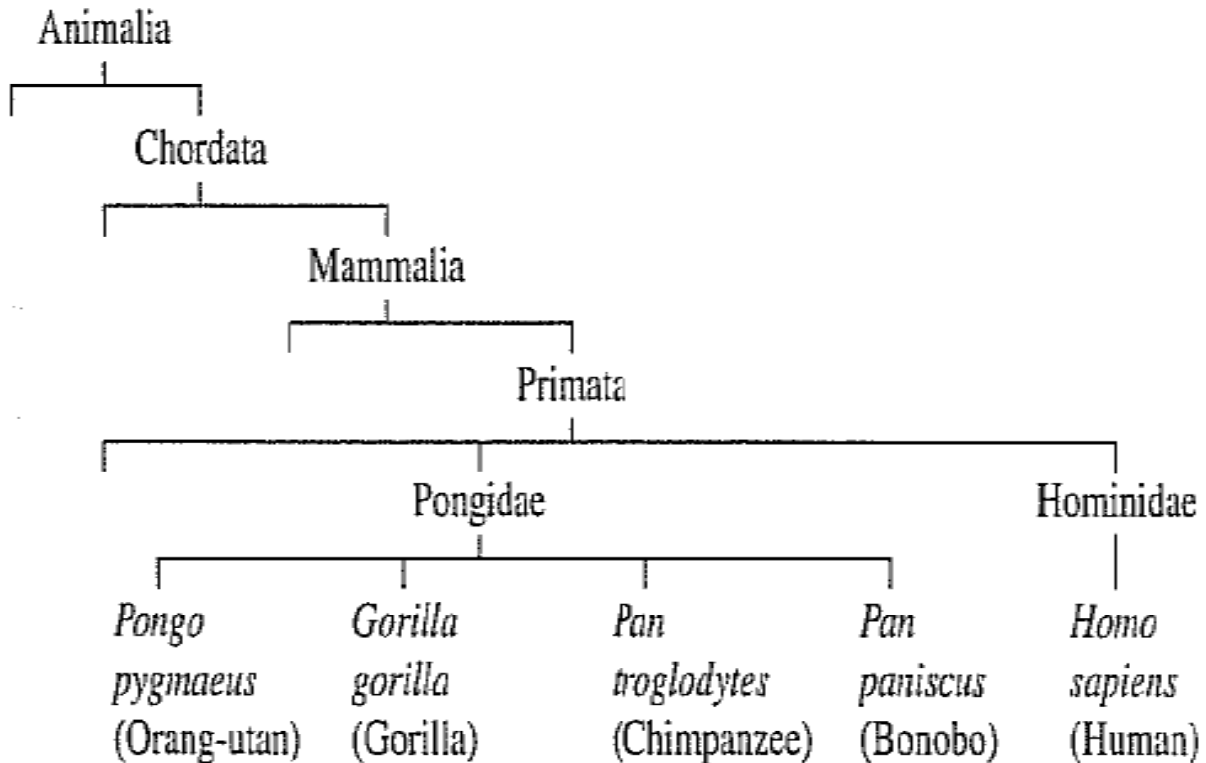


Figure 6.1. Classification of Some Primates

(a) Use the information from Figure 6.1,

(i) State which genus does the orang-utan belong? [1]

(ii) State which order does the chimpanzee belong? [1]

(b) Explain why this classification is both hierarchical as well as phylogenetic. [2]

The Galapagos are a very isolated group of islands. Species of birds called finches live on these islands. Scientists think that the different species have evolved from a population of one species which managed to reach the islands thousands of years ago from South American mainland. Scientists used base sequences in the DNA of the finches to work out the evolutionary relationship between the species. Figure 6.2 shows the evolutionary relationship.

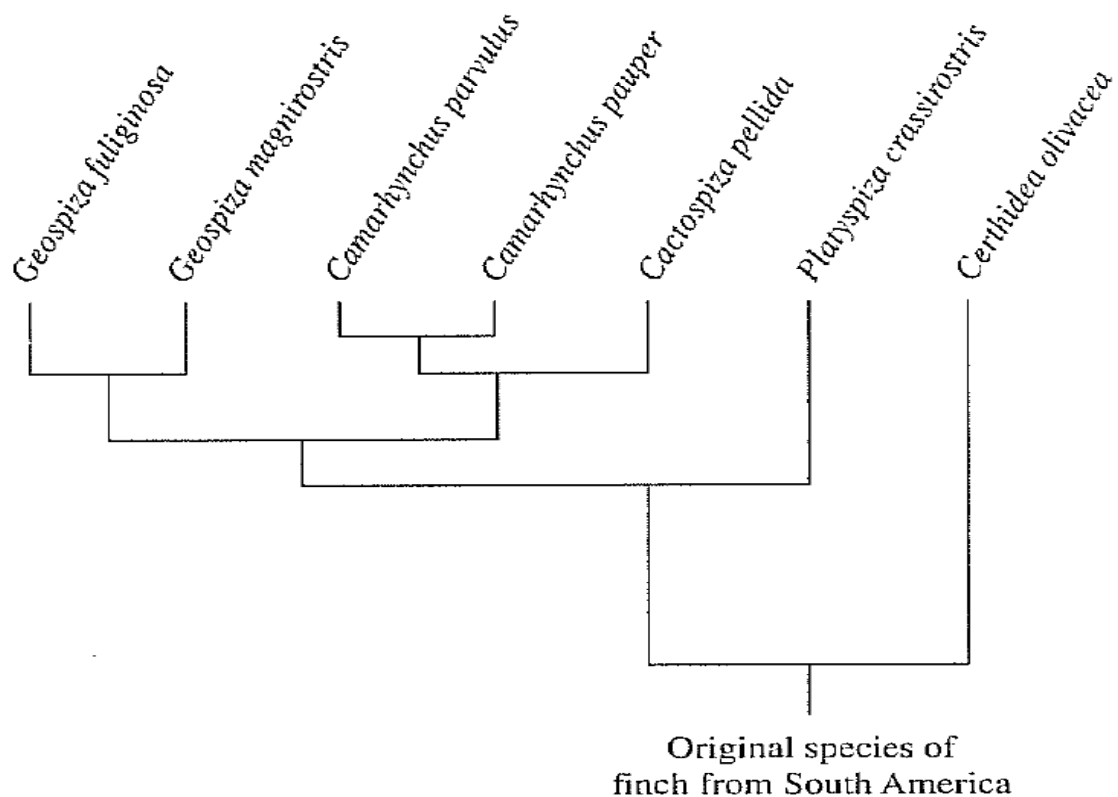


Figure 6.2. Evolutionary Relationship between the Finches

(c) Using the information in the diagram, state the present-day species of finch that evolved first on the Galapagos islands? Explain your answer. [2]

Finch species:

Explanation:

(d) Explain why evaluating the DNA base sequences can work out the evolutionary relationship between the species of finches. [2]

(e) The beak of each species is adapted for feeding on a particular type of food. *Geospiza fuliginosa* evolved as an eater of small seeds. *Geospiza magnirostris* evolved on a different island as an eater of large seeds.

(i) Suggest how these different species of finch might have evolved from one species. [3]

(ii) Describe how biogeography may provide support for the evolution of the different species of finch. [3]

Section B: Essay question

Please choose **EITHER** Question 7 **OR** Question 8.

Write your answers on the writing papers.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate. Your answers must be in continuous prose and must be set out in section (a), (b) and (c) as indicated in the question.

Either

Question 7

- a) Explain the roles of nucleic acids in the synthesis of proteins in a cell. [7]
- b) Distinguish between the regulation of protein synthesis in an animal cell with that of a bacterial cell. [8]
- c) Distinguish between gene mutations and chromosomal aberrations. [5]

Or

Question 8

- a) Distinguish between phospholipids and triglycerides stating clearly the features that allow each to serve different functions. [6]
- b) Describe how phenotypic ratios in genetic crosses can reveal about the type of gene interactions. [8]
- c) With reference to the stages of cell signaling, describe the differences between the response of liver cells to glucagon and insulin. [6]